# Perturbations: PING tolerates 80% drops but only 15% adds

exp037 · 30 May 2026 · Draft

The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained.

## Abstract

Perturbs the hidden spike stream of trained PING and COBA networks at inference (drop spikes, add spikes) to ask whether the PING rate floor is dynamical or informational. PING tolerates ≈ 80% drop but only ≈ 15% add before accuracy collapses; COBA is roughly flat to both perturbations. The asymmetry — drops forgiven, adds break the gating — is the gamma cycle made visible: a structural feature of the architecture, not a readout-side trade-off.

## Method

| Parameter | Value |
|---|---|
| Integration timestep $\Delta t$ | 0.1 ms |
| Trial duration $T$ | 200 ms |
| MNIST samples (80/20 stratified split of 2000) | 1600 train / 400 test (≈ 2.9% of the 70k-sample MNIST corpus) |
| Epochs | 10 |

The PING and COBA baseline definitions and training recipe are in exp025, where the rate-floor mechanism is also worked out. This entry tests whether the floor is **dynamical** (locked by the cycle period) or **informational** (locked by the readout's spike-count requirement) by perturbing the hidden spike stream of the trained networks at inference.

**The perturbation.** A per-step callback on the COBANet's *_hidden_perturb_fn* slot fires every timestep of every trial — no warm-up, no schedule, no exclusion. Trials are $T = 200$ ms at $\Delta t = 0.1$ ms → 2000 fires per trial; the test set (≈ 400 trials) gives ≈ 800 000 fires per (model, mode, level) sweep point against the same trained network. At each timestep $t$:

1. *update conductances* from the previous step's spikes: $g_e^{(t)} \leftarrow g_e^{(t-1)} \cdot d_{\text{AMPA}} + \tilde{\boldsymbol{s}}^{E,(t-1)} W_{ee} + \text{input}^{(t)} W_{\text{in}}$, with the analogous update for $g_i^{(t)}$ from $\tilde{\boldsymbol{s}}^{I,(t-1)} W_{ie}$ and the E→I conductance from $\tilde{\boldsymbol{s}}^{E,(t-1)} W_{ei}$;
2. *LIF step* integrates $V^{(t)}$ and emits raw spike vectors $\boldsymbol{s}^{E,(t)} \in \{0,1\}^{B\times N_E}$, $\boldsymbol{s}^{I,(t)} \in \{0,1\}^{B\times N_I}$;
3. *perturbation callback* rewrites the raw vectors → $\tilde{\boldsymbol{s}}^{E,(t)}, \tilde{\boldsymbol{s}}^{I,(t)}$;
4. *record* the perturbed vectors into the spike buffer;
5. *readout accumulator* adds $\tilde{\boldsymbol{s}}^{E,(t)} W_{\text{out}}$ to the mem-mean integrator.

Step 1 of timestep $t+1$ consumes the perturbed spikes through $W_{ee}, W_{ei}, W_{ie}$: a dropped E spike fails to drive I next step and contributes nothing to the readout; an injected I spike adds inhibition next step and counts in the rate metric. E and I get the same mode and level with independent draws.

**Drop mode.** For each (batch, neuron, timestep) slot $i$, draw $u_i \sim \text{Uniform}(0,1)$ i.i.d. and keep each emitted spike with probability $1 - p_{\text{drop}}$:

$$\tilde{s}_i = s_i \cdot \mathbb{1}\left[u_i \geq p_{\text{drop}}\right].$$

Drop thins the count fed to both the readout and the next-step conductance update while leaving the E→I→E loop intact. Sweep $p_{\text{drop}} \in \{0.0, 0.1, ..., 1.0\}$, 11 levels.

**Add mode.** For each slot draw $u_i \sim \text{Uniform}(0, 1)$ i.i.d. and flip silent slots on at Poisson statistics, phase-independent:

$$\tilde{s}_i = \min(s_i + \mathbb{1}[u_i < r_{\text{add}} \cdot \Delta t/1000], 1).$$

Sweep $r_{\text{add}} \in \{0, 2, ..., 40\}$ Hz, 21 levels. On the right panel of Figure 1 the add level is expressed **per population** — $r^E_{\text{add}} = \text{pct} \cdot \bar{r}_E$, $r^I_{\text{add}} = \text{pct} \cdot \bar{r}_I$, each architecture scaled by its own baseline — because the baselines differ by an order of magnitude (COBA $\bar{r}_E = 65$ Hz vs PING $\bar{r}_E = 6$ Hz), so a fixed "10 Hz of noise" would be 15% of COBA's rate but 170% of PING's. The per-step RNG is seeded separately from the input encoder, so the Poisson input stream matches the unperturbed baseline. Total: 2 models × (11 drop + 21 add) = 64 forward passes.

## Results

pending re-run with new canonical data

Figure 1: **Left (drop)**: Bernoulli mask, % of emitted spikes dropped. **Right (add)**: Poisson injection per population (each scaled by its own baseline, so the comparison is architecture-fair). The asymmetry is direct: PING tolerates drop up to ≈ 80% but collapses on add between 15% and 50% relative noise, while COBA stays at 75% even with 100% extra Poisson.

pending re-run with new canonical data

Figure 2: Trained PING replayed on the same MNIST digit 0 trial across the six drop levels of Figure 1′s left panel (0, 30, 60, 80, 90, 100%); E (black) above I (red). The gamma cadence persists across all sub-total levels — dropped spikes thin the train without injecting phase-incoherent activity.

pending re-run with new canonical data

Figure 3: Trained PING replayed across the six pct-of-baseline add levels of Figure 1′s right panel (0, 5, 15, 25, 40, 80%). The cycle visibly dissolves across 25–40%, matching Figure 1′s accuracy cliff.

pending re-run with new canonical data

Figure 4: Trained COBA replayed across the same drop sweep as Figure 2. With no cycle to preserve, dropping spikes just thins a uniform asynchronous mean.

pending re-run with new canonical data

Figure 5: Trained COBA replayed across the same pct-of-baseline add sweep as Figure 3. Added spikes blend into COBA's asynchronous mean — no temporal structure to corrupt, just a higher mean rate.